Attack on the Final Planar Radialization Step

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1 Setup and target

Let $E = E_{\hat{t}} \subset \mathbb{R}^2$ be the $\sqrt{\delta}$-stopped level. Normalize $|E| = \pi + O(\sqrt{\delta})$, and write after harmless rescaling

$$|E| = \pi.$$

The available inputs are

$$\delta_T(E) \le C\delta, \qquad D_I(E) \le \varepsilon, \qquad D_H(E) \le \varepsilon, \qquad \varepsilon = C_K\sqrt{\delta}, \tag{1.1}$$

and, from the selected-class interpolation,

$$\big\| |\nabla u_E| - c_E \big\|_{L^\infty(\partial E)} \le \eta_K, \qquad \eta_K \to 0 \quad (K \to \infty). \tag{1.2}$$

We also use the uniform Lipschitz character of the stopped level, with constant L, and a bounded ambient diameter.

The needed conclusion is a small-amplitude radial graph G such that

$$|E\Delta G| \le C_K\sqrt{\delta}, \qquad \delta_T(G) \le C_K\delta. \tag{1.3}$$

Once (1.3) holds, the direct radial-graph theorem gives $\mathcal{A}(G)^2 \le C\delta_T(G) \le C_K\delta$, and the transfer back to E and then to U closes the desired estimate.

2 Do not convexify everything

The previous convex-envelope idea is useful but too crude if applied to all of E. A wide shallow radial indentation can have area $O(\sqrt{\delta})$ and perimeter excess $O(\sqrt{\delta})$, while already being a perfectly good radial graph. Convexifying it would create a volume error too large for the torsion-deficit bookkeeping.

Therefore the repair should fill only the genuinely non-radial pockets: holes, satellites, folds, same-ray missing intervals, and overhangs. Radial indentations with one crossing per ray should be left untouched.

3 Components are ruled out by Serrin, not just by D_I

The D_I estimate already gives satellite area $O(\varepsilon^2)$. In the selected Serrin situation one can say more.

Lemma 3.1 (No small Serrin components)**.** *Assume $|E| = \pi$, $P(E) \le 2\pi + C\varepsilon$, and*

$$c_0 \le |\nabla u_E| \le c_1 \quad \text{on } \partial E$$

with c_0 sufficiently close to $1/2$. Then, for ε small, E has only one connected component.

Proof. On every connected component E_j, the restriction of u_E is the torsion function of E_j. The divergence theorem gives

$$|E_j| = \int_{\partial E_j} |\nabla u_E| \, d\mathcal{H}^1 \geq c_0 P(E_j). \tag{3.1}$$

The planar isoperimetric inequality gives

$$P(E_j) \geq 2\sqrt{\pi |E_j|}.$$

If $|E_j| > 0$, then

$$|E_j|^{1/2} \geq 2c_0\sqrt{\pi}.$$

For c_0 close to $1/2$, each nontrivial component has area close to π. Since the total area is π, there can be only one component for ε and η_K small. □

Remark 3.2. *This is a useful improvement over pure D_I. Tiny satellites are allowed by small perimeter deficit, but they are incompatible with a pointwise Serrin lower bound at the ball scale.*

4 Closed holes are harmless

Let p_h be the total perimeter of bounded hole boundaries. Since holes contribute to $P(E) - 2\pi$, $p_h \leq C\varepsilon$. By the planar isoperimetric inequality,

$$|H_{\text{holes}}| \leq \frac{p_h^2}{4\pi} \leq C\varepsilon^2 \leq C_K \delta. \tag{4.1}$$

Filling them changes the volume by $O(\delta)$. The monotone fill-and-rescale lemma from the earlier planar radialization note then gives

$$\delta_T(E^\sharp_{\text{filled}}) \leq C(\delta_T(E) + |H_{\text{holes}}|) \leq C_K \delta. \tag{4.2}$$

5 The planar GMT pocket lemma

Choose the eventual center z from the Brandolini/Plan-2 same-center package. The geometric input needed at this stage is not merely that ∂E is Lipschitz. One also needs the already advertised annular core information, namely

$$B_{r_-}(z) \subset E \subset B_{r_+}(z), \qquad r_- = 1 - \eta, \quad r_+ = 1 + \eta, \tag{5.1}$$

with η smaller than the fixed graph-entry threshold. Closed holes may first be filled by Section 4; this changes area by $O(\varepsilon^2)$, so it is harmless for the present estimate.

For a.e. $\theta \in S^1$, write

$$E_\theta = \{r > 0 : z + re^{i\theta} \in E\}.$$

Since $B_{r_-}(z) \subset E$, the slice E_θ starts at the origin. Define the radial hull

$$R(\theta) := \operatorname{ess\,sup} E_\theta, \qquad G := \{z + re^{i\theta} : 0 < r < R(\theta)\}. \tag{5.2}$$

Thus one-crossing radial indentations are left unchanged. The set $G \setminus E$ consists exactly of same-ray gaps lying behind some outer piece of E.

Lemma 5.1 (Radial-hull boundary decomposition)**.** *Let $E \subset \mathbb{R}^2$ be a bounded connected Lipschitz domain satisfying* (5.1)*. Let G be defined by* (5.2)*, and let $\{Q_j\}$ be the connected components of $G \setminus E$. Then G has finite perimeter and*

$$P(G) + \sum_j P(Q_j) \leq P(E). \tag{5.3}$$

Proof. Translate z to the origin. The polar map

$$\Phi : S^1 \times (r_-, r_+) \to B_{r_+} \setminus \overline{B_{r_-}}, \qquad \Phi(\theta, r) = re^{i\theta},$$

is bi-Lipschitz, so rectifiability and perimeter identities may be checked in the cylinder. For a.e. θ, the one-dimensional slice E_θ is a finite-perimeter subset of $(0, \infty)$, and because of the core inclusion its reduced boundary consists of an odd ordered family

$$r_1(\theta) < r_2(\theta) < \cdots < r_{2m(\theta)+1}(\theta).$$

The outer endpoint $r_{2m+1}(\theta)$ is $R(\theta)$; the earlier endpoints are precisely the two sides of the radial gaps that form $G \setminus E$.

We now apply the area formula, equivalently the BV graph-perimeter formula, to the countably many polar sheets of ∂E. On a smooth sheet $\theta \mapsto r(\theta)$, the Euclidean length element is

$$ds = \sqrt{r^2\, d\theta^2 + dr^2}.$$

The same formula, with the usual vertical jump contribution, holds for BV polar graphs. The perimeter of G is obtained by selecting, for each θ, the outermost endpoint $R(\theta)$. If this selected endpoint jumps, the vertical jump segment in ∂G is exactly the jump term of the corresponding outer sheet in the BV graph formula.

The components Q_j are the gaps between consecutive endpoint functions $r_{2i-1}(\theta)$ and $r_{2i}(\theta)$. Their perimeter is bounded by the BV graph perimeters of these two endpoint sheets, including the vertical jump terms at the angles where the gap opens or closes. Consequently the perimeter of G, plus the perimeters of all pockets Q_j, is bounded by the sum of the polar graph perimeters of all sheets of ∂E. By the area formula in the bi-Lipschitz polar chart, this sum is $P(E)$, giving (5.3). ☐

Theorem 5.2 (Lipschitz pocket lemma)**.** *Let $E \subset \mathbb{R}^2$ be a bounded connected Lipschitz domain with $|E| = \pi$,*

$$P(E) \leq 2\pi + \varepsilon,$$

and assume the annular core condition (5.1)*. Let G be the radial hull* (5.2)*. Then G is a radial graph,*

$$|G \setminus E| \leq \frac{\varepsilon^2}{4\pi}. \tag{5.4}$$

In particular, if $\varepsilon = C_K\sqrt{\delta}$, then

$$|G \setminus E| \leq C_K \delta. \tag{5.5}$$

Proof. Write

$$v := |G \setminus E|, \qquad h := \sum_j P(Q_j).$$

Since $E \subset G$, $|G| = \pi + v$. By Lemma 5.1 and the isoperimetric inequality applied to G,

$$2\pi + \varepsilon \geq P(E) \geq P(G) + h \geq 2\sqrt{\pi(\pi + v)} + h \geq 2\pi + h. \tag{5.6}$$

Thus $h \leq \varepsilon$. Applying the planar isoperimetric inequality to each pocket Q_j gives

$$|Q_j| \leq \frac{P(Q_j)^2}{4\pi}.$$

Summing over j,

$$v = \sum_j |Q_j| \leq \frac{1}{4\pi} \sum_j P(Q_j)^2 \leq \frac{1}{4\pi} \Big(\sum_j P(Q_j)\Big)^2 \leq \frac{\varepsilon^2}{4\pi}.$$

This proves (5.4). The inclusion (5.1) also gives

$$B_{r_-}(z) \subset G \subset B_{r_+}(z),$$

so after the harmless area normalization G is a small-amplitude radial graph whenever η is small. $\square$

Remark 5.3. *The theorem is deliberately about the radial hull, not the convex hull. A wide shallow one-crossing indentation has no same-ray gap and is not filled. Thus the argument avoids the earlier $O(\sqrt{\delta})$-volume loss.*

6 Closure using the pocket lemma

Fill the closed holes from Section 4 and then take the radial hull (5.2). Call the result G. The total filled volume is

$$|G \setminus E| \le C_K \delta. \tag{6.1}$$

The monotone fill-and-rescale lemma yields

$$\delta_T(G^\sharp) \le C\big(\delta_T(E) + |G \setminus E|\big) \le C_K \delta. \tag{6.2}$$

The annular amplitude required by the direct radial-graph theorem is exactly (5.1). Therefore $G^\sharp$ satisfies (1.3), and the direct radial-graph theorem gives

$$\mathcal{A}(G^\sharp)^2 \le C\, \delta_T(G^\sharp) \le C_K \delta. \tag{6.3}$$

Since $|E \Delta G^\sharp| \le C_K \sqrt{\delta}$ after area normalization, the usual transfer gives $\mathcal{A}(E) \le C_K \sqrt{\delta}$, and then Step 5 transfers the estimate back to U.

7 What remains outside this GMT step

The GMT pocket step is proved above for Lipschitz boundaries. The remaining non-GMT inputs needed before applying it are:

1. the $\sqrt{\delta}$-stopped level;
2. exact deficit transfer $\delta_T(E) \le C\delta$;
3. pointwise Serrin control on ∂E;
4. no small components from the Serrin lower bound;
5. the annular core condition (5.1) with the same center.

Thus the old "pocket area" obstruction is no longer the bottleneck: once the level is known to have the fixed annular core, radializing its same-ray gaps costs $O((P(E) - 2\pi)^2) = O_K(\delta)$, which is the required torsion scale.